BayScen: Standards-Grounded Scenario Generation for Autonomous Vehicle Testing via Bayesian Networks and Capability Abstraction

***Abstract*—Scenario-based testing of autonomous vehicles faces a fundamental trilemma: combinatorial testing achieves coverage but generates physically implausible scenarios; search-based methods find critical failures but violate naturalistic driving dependencies; data-driven methods maintain realism but cannot escape their training distribution. This paper introduces BayScen, which navigates this trilemma through two mechanisms. First, an ISO 34503:2023-grounded capability abstraction collapses millions of raw environmental parameter combinations into three distinct degradation regimes—sensor perception, surface traction, and lateral stability—making exhaustive combinatorial coverage tractable while preserving safety-relevant distinctions. Second, a Bayesian Network learned from real-world observations encodes probabilistic dependencies among environmental parameters, so generated scenarios respect the statistical regularities of naturalistic driving. Across three NHTSA pre-crash scenarios, two ADS architectures, and six baselines: while adversarial search produces more raw collisions, 91–95% violate physical constraints; BayScen's critical scenarios are $\times 7$ more likely to be physically plausible (40% vs. 4–7%) and achieve the highest ODD coverage by 12–17 percentage points. Beyond aggregate metrics, capability-organized failure profiling reveals architecture-specific vulnerability patterns—perception convergence under extreme degradation, traction discrimination between architectures—invisible to methods lacking this abstraction.**

***Index Terms*—autonomous vehicle testing, scenario generation, safety-critical systems, operational design domain, combinatorial testing, bayesian networks.**

I. Introduction

Validating the safety of autonomous vehicles (AVs) is fundamentally a rare-event problem: achieving statistical confidence in reliability would require driving billions of kilometers [1]—an economic and temporal impossibility. Scenario-based testing (SBT) addresses this by decomposing the operational design domain (ODD) into structured, reproducible test cases [2], and is increasingly mandated by ISO 21448 (SOTIF) [3] and ISO 34503 [4].

Yet the raw parameter space of a scenario is not flat. Environmental parameters do not act on an AV independently — they act *through* a small set of safety-critical capabilities they degrade. This structure is the key insight behind **BayScen**:by abstracting at the level of capabilities rather than raw parameters, BayScen collapses a space of millions of combinations into a compact set of genuinely distinct degradation regimes, making exhaustive combinatorial testing tractable without discarding any safety-relevant distinction.

This insight navigates what is otherwise a structural trilemma in scenario-based testing. *Coverage* demands broad ODD exploration. *Realism* demands that scenarios respect the probabilistic dependencies of naturalistic driving. *Criticality* demands concentration on the rare, dangerous tail of the operational envelope. Existing methods satisfy at most one of these simultaneously. Combinatorial testing [5, 6] guarantees structural coverage but treats all parameter combinations as equally probable, wasting budget on scenarios that could never occur in real-world driving. Search-based methods [7, 8, 9] efficiently find failures but frequently violate naturalistic dependencies, producing physically implausible conditions [10, 11] — a claim we validate empirically by benchmarking against AvFuzzer [12]. Data-driven approaches [13, 14, 15] maintain behavioral realism but are trapped by a coverage paradox: they can only reproduce scenarios resembling their training distribution, systematically underrepresenting the rare situations that matter most for safety [16].

BayScen navigates this trilemma through two principled mechanisms. First, an ISO 34503:2023-grounded capability abstraction — the same three variables (sensor perception, surface traction, lateral stability) applied unchanged across all scenario types — provides exhaustive, scenario-agnostic ODD coverage. Second, a Bayesian Network (BN) learned from over 43,000 real-world weather observations encodes probabilistic dependencies among environmental parameters, embedding realism in generation rather than as a post-hoc filter; rarity-prioritized sampling over the BN posterior then concentrates the suite on the tail conditions most likely to expose system vulnerabilities.

We evaluate BayScen in CARLA [17] across three NHTSA Pre-Crash scenarios [18] spanning five variations, two actor types, and two conflict geometries, against two contrasting ADS architectures and six baselines including AvFuzzer as the strongest adversarial comparator. An ablation baseline (BayScen-Common) isolates the contribution of rarity prioritization.

The contributions of this paper are:

1) **BayScen: an ISO-grounded scenario generation framework** that integrates Bayesian Networks with ISO 34503-grounded capability abstraction and combinatorial testing, balancing coverage, realism, and criticality rather than trading one against the others.
2) **Scenario-agnostic capability abstraction.** Three capability variables derived from ISO 34503:2023 apply unchanged across all evaluated scenario types, positioning the abstraction as a reusable foundation for ODD-driven test design beyond the specific scenarios evaluated here.

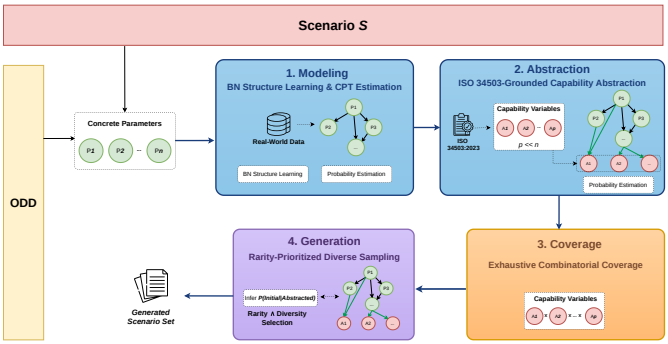


Fig. 1: Overview of BayScen's four-stage pipeline. **(1) Modeling** learns a Bayesian Network from real-world driving data, encoding probabilistic dependencies among environmental parameters. **(2) Abstraction** collapses n raw parameters into $p \ll n$ ISO 34503:2023-grounded capability variables, each grouping parameters by the ADS capability they degrade. **(3) Coverage** exhaustively enumerates all capability configurations via combinatorial testing. **(4) Generation** samples concrete scenarios from the conditional BN posterior, prioritizing rare and diverse combinations to target the safety-critical tail of the driving distribution.

3) **Comprehensive empirical evaluation** across three NHTSA scenarios, two ADS architectures, and six baselines. BayScen's critical scenarios are $\times 7$ more likely to be physically plausible than the strongest adversarial baseline (40% vs. 4–7%), and it leads all methods on ODD coverage by 12–17 percentage points. Capability-organized failure profiling further reveals architecture-specific vulnerability patterns invisible to aggregate evaluation.

The remainder of this paper is organized as follows. Section II describes the BayScen framework. Section III presents the experimental setup. Section IV reports the results. Section V discusses related work. Section VI addresses threats to validity. Section VII concludes.

II. Approach

A. Overview

BayScen generates concrete test scenarios from two inputs: the system's ODD and the functional scenario under test. The ODD defines the admissible envelope of environmental and operational conditions; the functional scenario defines the specific situation to be exercised, including road layout, actors, maneuvers, and trajectories. Together they yield the logical scenario from which concrete test cases are instantiated [19].

BayScen addresses the three challenges of scenario-based testing—coverage, realism, and criticality—through four tightly coupled stages (Figure 1): *Modeling*, *Abstraction*, *Coverage*, and *Generation*. Abstraction and Coverage make exhaustive combinatorial testing tractable; Modeling keeps generated scenarios naturalistic; and Generation's rarity- and diversity-aware sampling concentrates the suite on the safety-critical tail. The following subsections detail each stage.

B. Modeling the Scenario Space as a Bayesian Network

We model the parameter space as a probabilistic graphical model that captures the joint distribution over scenario variables and their causal dependencies. The BN serves two roles in BayScen: it is the source of realism (generated scenarios inherit real-world statistical regularities) and the inference engine for scenario instantiation (conditioning on capability configurations yields naturalistic concrete scenarios).

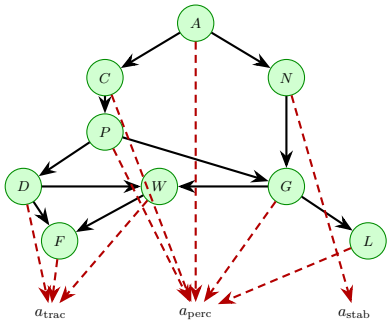


Fig. 2: Learned Bayesian Network (S2/S3 shown). **Green** nodes are environmental parameters with learned structural dependencies (**black solid** edges); **leaf** nodes are the three ISO 34503-grounded capability variables, linked by deterministic aggregation edges (**red dashed**; §II-C). *Environmental codes:* A sun altitude, C cloudiness, N wind intensity, P precipitation, D precipitation deposits, W wetness, G fog density, L fog distance, F road friction. *Capability codes:* a_{perc} perception, a_{trac} traction, a_{stab} stability. S1 omits node A and its edges (used only in S2/S3).

1) Dataset Preparation

We use real-world observational data comprising environmental and road condition parameters (weather, visibility, road surface, and related conditions). Intersection parameters (vehicle start and goal locations) are modeled deterministically to ensure trajectory conflicts at the intended collision points. The dataset captures the natural co-occurrence dependencies between meteorological phenomena and road surface states that constitute the realism signal exploited in the generation stage.

2) BN Structure Learning

We encode established domain knowledge (e.g., precipitation causes wetness; fog reduces visibility) as *soft ancestral constraints*—directed paths that are *encouraged but not enforced* during structure search—and learn the network with the error-tolerant, knowledge-guided framework of Ban et al. [20]. Because the constraints enter the scoring function as soft penalties, observational data that strongly contradicts a prior overrides it, so the learned structure reflects the empirical dependencies rather than the assumed ones. We use Bi-CaMML [21], which natively supports soft ancestral constraints, to obtain the final structure (Fig. 2). The same structure is reused across all three scenarios; Scenario 1 omits the sun-altitude node (and its edges), which applies only to the cyclist (S2) and cut-in (S3) scenarios. The network comprises nine environmental-parameter nodes with 11 learned structural edges, augmented in the capability-abstraction stage (§II-C) by three capability leaf nodes (one per degradation pathway) and nine aggregation edges.

3) BN Parameterization

With the structure fixed, we estimate the conditional probability distributions using Bayesian parameter estimation with a BDeu prior [22]. Bayesian smoothing (equivalent sample size $= 5$ [23]) ensures well-defined estimates even for rare parameter combinations, yielding smooth distributions that support reliable sampling throughout the tail of the joint distribution.

C. ODD-Grounded Capability Abstraction

1) Why Abstract at the Capability Level

A raw environmental parameter space of $n{=}8$ variables at $K{=}6$ values each spans $6^8 \approx 1.7 \times 10^6$ combinations—intractable for systematic simulation-based testing. Yet this count substantially overestimates the number of qualitatively distinct conditions: raw parameters do not act on the AV independently, but through the functional capabilities they degrade. Multiple parameters can jointly saturate the same capability, rendering many concrete combinations indistinguishable from a safety perspective. Abstracting at the capability level therefore collapses this redundancy, reducing the effective test space to a compact set of genuinely distinct degradation regimes and concentrating the test budget on the distinctions that matter for safety.

2) Environmental Parameters

We use CARLA [17], an open-source urban driving simulator widely adopted in AV testing research for its high-fidelity sensor models and fine-grained environmental control. We include all nine of its controllable environmental parameters—fog density, fog distance, cloudiness, precipitation, precipitation deposits, road friction, wetness, wind intensity, and sun altitude angle (Table I)—so that the parameter space is exhaustive by construction rather than a subjective selection. The BN is learned from 43,000+ hourly observations from the Norwegian Meteorological Institute's Frost API [24] (Dec. 2020–Nov. 2025), covering fog, precipitation, wind, cloudiness, and road-surface conditions directly mappable to these parameters.

3) ISO-Grounded Capability Variables

ISO 34503:2023 defines the attribute classes that *motivate* our grouping; the specific three-variable operationalization is a methodological design choice whose validity we establish empirically (RQ4). Among its three ODD attribute classes (i.e., *scenery elements*, *environmental conditions*, and *dynamic elements*) [4], environmental conditions are uniquely suited as the abstraction target. Scenery elements are map-fixed and do not vary within a test campaign. Dynamic elements are governed by the functional scenario definition. Environmental conditions, by contrast, vary continuously, interact through physical causal chains, and degrade AV performance through mechanisms that ISO 34503 explicitly enumerates. Crucially, they are *scenario-agnostic*, making a single abstraction reusable across all scenario types.

We organize the enumerated ISO mechanisms into three capability pathways, each realized as one leaf node in the BN with directed edges from its contributing concrete parameters (full mapping in Table I): *(i) sensor perception* (a_{perc}), from atmospheric particulates and the resulting obscuration/MOR (§10.3) and from illumination governed by solar elevation and cloud cover (§10.4)—five parameters: fog density, fog distance, cloudiness, sun altitude angle, and precipitation (the last included here as a modeling choice); *(ii) surface traction* (a_{trac}), from the induced surface conditions of §9.3.7 (icy, flooded, standing water, snow, wet, contamination)—three parameters: road friction, wetness, and precipitation deposits; and *(iii) lateral stability* (a_{stab}), from cross-wind loading (§10.2.3)—wind intensity alone. Although §9.3.7 is taxonomically under scenery elements, its induced surface conditions are consistent with the §8.2 effect statement that environmental conditions can affect manoeuvrability through changing drivable-area surface conditions [4], making their joint inclusion physically coherent. Each capability variable is treated as causally downstream of its contributing concrete parameters.

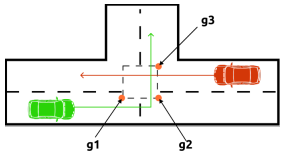


Fig. 3: Non-signalized junction configuration showing the three trajectory intersection points (g1, g2, g3) where ego vehicle (green) and adversarial agent (red) trajectories intersect.

4) Aggregation and Discretization

Each capability variable's CPD is computed by uniform-weight aggregation over its parent parameters: within a group the parameters share the same *effect target*, and their pairwise causal relationships are already captured by the BN structure over the concrete variables, so the leaf represents only their *combined* effect. Absent reliable data on relative influence, uniform weights are a neutral maximum-entropy choice that avoids biasing the abstraction toward any one parameter.

All variables share a uniform $K{=}6$ discretization: a finer resolution would spend budget on parameter differences with little effect on ADS capability, while a coarser one would merge materially distinct regimes. $K{=}6$ also aligns with ISO 34503:2023, which enumerates six induced surface conditions (§9.3.7) and six precipitation intensity levels (§10.2.4), making it a standard-aligned resolution rather than an arbitrary hyperparameter.

Each leaf's CPD concentrates probability mass on the value matching its parents' average; conditioning on a target capability value a^* then induces a well-defined posterior $P(X \mid A = a^*)$ from which concrete parameter assignments are drawn by ancestral sampling.

5) Scenario-Level Abstraction: Conflict Geometry

The three ODD-level capability variables are scenario-agnostic and apply unchanged across all functional scenarios evaluated in this paper. Junction scenarios additionally present a trajectory space that can be compressed by geometric reasoning.

With 3 start and 3 goal positions per agent, the ego and adversarial trajectory space contains $3^4{=}81$ combinations, the large majority of which yield non-conflicting paths irrelevant to safety testing. Following Tahir and Alexander [25], we map all start-goal configurations that produce geometrically equivalent trajectory intersections to a single *Conflict Geometry* state g. The mapping is deterministic from road topology: each configuration yields exactly one of three distinct conflict points (g_1, g_2, g_3, Figure 3), reducing 81 combinations to 3 geometrically distinct states. This variable is scenario-specific and does not apply to the cut-in scenario, where the binary direction variable (left/right) is already fully expressive and requires no further abstraction.

The full abstracted variable set thus decomposes into two levels:

$$\mathcal{A} = \underbrace{\{a_{\text{perc}},\ a_{\text{trac}},\ a_{\text{stab}}\}}_{\text{ODD-level: scenario-agnostic}} \cup \underbrace{\{g\}}_{\text{scenario-level: junction-specific}} \tag{1}$$

D. Coverage: Combinatorial Testing on Capability Variables

The capability abstraction reduces the combinatorial space from millions of raw parameter combinations to a compact, structured set of capability configurations. The coverage phase exploits this structure by exhaustively enumerating all combinations of capability variable values, guaranteeing that every distinct degradation regime is exercised at least once.

Formally, let each capability variable $a_i \in \mathcal{A}$ take values in a discrete set $\mathcal{V}_i$. The full combinatorial test suite is:

$$\mathcal{T} =_{a_i \in \mathcal{A}} \mathcal{V}_i \tag{2}$$

Each configuration $t \in \mathcal{T}$ specifies a complete instantiation of all capability variables, defining a target degradation condition that the system under test must face. Because the abstraction has already collapsed behaviorally equivalent parameter combinations, every configuration in $\mathcal{T}$ is genuinely distinct from a safety perspective—exhaustive coverage here is both tractable and meaningful. For scenario spaces where even the abstracted space is large, t-way covering arrays can be substituted for $\mathcal{T}$ without modifying any other stage of the framework.

E. Generation: Sampling from the Bayesian Network

The generation phase transforms each capability configuration $t \in \mathcal{T}$ into a concrete, executable scenario. The BN is the mechanism that makes this transformation principled: because capability variables are leaf nodes whose parents are the concrete parameters, conditioning on t induces a well-defined posterior distribution over the concrete space that simultaneously satisfies the target capability conditions *and* respects the naturalistic dependencies learned from real-world data.

1) Conditional Sampling

For a target configuration $t = (a_1 = v_1^*,\ a_2 = v_2^*,\ \ldots,\ a_m = v_m^*)$, we sample concrete parameter assignments from:

$$P(\mathcal{X} \mid a_1 = v_1^*,\ \ldots,\ a_m = v_m^*) \tag{3}$$

This posterior is non-trivial: it concentrates probability mass on concrete parameter combinations that jointly produce the target capability levels while remaining consistent with the correlations in the learned BN structure. In this sense the BN acts as a *soft prior* over parameter co-occurrence: it biases sampling toward jointly realistic combinations rather than hard-filtering the space, and no candidate scenario is excluded a priori. Scenarios drawn from this distribution therefore tend toward realism—they inherit the statistical regularities of naturalistic driving conditions rather than being assembled from independently sampled parameters.

We draw N=100,000 candidates per capability configuration from the conditional BN distribution—large enough that every non-negligible tail combination is present in the pool before rarity ranking is applied.[1]

Algorithm 1 Rarity-Prioritized Diverse Scenario Generation

Require: Bayesian Network BN, capability configurations $\mathcal{T}$
Ensure: Scenario set $\mathcal{S}$: exhaustive in capability coverage, rare and diverse in concrete parameters

$\mathcal{S} \leftarrow \emptyset$
for each configuration $t \in \mathcal{T}$ **do**
  $\mathcal{C} \leftarrow \textsc{SampleRare}(BN, t)$ ▷ Sample pool from $P(\mathcal{X} \mid t)$, rank by rarity, retain rare candidates
  if $\mathcal{S} = \emptyset$ **then**
    $s^* \leftarrow \arg\min_{s \in \mathcal{C}}\ P(s \mid t)$ ▷ First scenario: rarest candidate
  else
    $s^* \leftarrow \arg\max_{s \in \mathcal{C}}\ \min_{s' \in \mathcal{S}}\ d(s, s')$ ▷ Max-min diversity among rare candidates
  end if
  $\mathcal{S} \leftarrow \mathcal{S} \cup \{s^*\}$
end for
return $\mathcal{S}$

2) Rarity-Prioritized, Diversity-Aware Selection

We apply a two-criteria selection strategy over a pool of candidates drawn from the conditional distribution.

a) Rarity.

Safety-critical events are rare in naturalistic driving data [16]: naive sampling from the conditional distribution would predominantly yield common parameter combinations that are already well-represented in AV training data, offering limited additional value for safety testing. Candidates are ranked by their empirical probability under the conditional distribution, and the rarest configurations are prioritized. Rare combinations correspond to the tail of the joint distribution over concrete parameters—they represent unusual but physically valid conditions that stress the system in ways common scenarios cannot. Crucially, rarity is assessed *within* the conditional distribution: candidates are rare relative to the target capability configuration, not globally, ensuring they still satisfy t.

b) Diversity.

From the pool of rare candidates, the final scenario is selected to maximize its minimum distance to all previously generated scenarios:

$$s^* = \arg\max_{s \in \mathcal{C}}\ \min_{s' \in \mathcal{S}}\ d(s,\ s') \tag{4}$$

where $d(s, s')$ is the normalized Euclidean distance over the concrete parameter space. This max-min criterion ensures that each new scenario occupies a region of the parameter space not already covered by the existing set, producing a final suite that is both rare and non-redundant.

c) Complete procedure.

Algorithm 1 presents the full generation loop. For each capability configuration t, a pool of candidate scenarios is sampled from the conditional BN distribution, ranked by rarity, and the most diverse rare candidate is selected. The process iterates over all $t \in \mathcal{T}$, accumulating a scenario set $\mathcal{S}$ that jointly achieves exhaustive capability coverage, naturalistic realism, and maximal parameter diversity.

[1]For any fixed combination with conditional probability $p \geq 10^{-4}$, the probability of its absence from the pool is at most $(1-p)^N \leq e^{-Np} \leq e^{-10} \approx 4.5{\times}10^{-5}$, so it is individually present with probability exceeding $1 - 4.5{\times}10^{-5}$.

TABLE I: Concrete parameters, their ISO 34503:2023 grounding, and the capability variable they contribute to. Parameters within a group are aggregated with equal weights in the BN leaf node CPD.

Concrete parameter	ISO 34503 clause	Safety-relevant effect	Capability variable
Sensor Perception (a_{perc}) — ISO §10.2.4, §10.3, §10.4			
Fog density, fog distance	§10.3 Particulates	Atmospheric obscuration (MOR)	a_{perc}
Cloudiness	§10.4(c) Illumination	Cloud cover affecting illumination	a_{perc}
Precipitation	§10.2.4 Precipitation	Optical perception degradation	a_{perc}
Sun altitude angle†	§10.4(d) Illumination	Solar elevation / lighting	a_{perc}
Surface Traction (a_{trac}) — ISO §9.3.7			
Road friction	§9.3.7 Surface cond.	Baseline friction coefficient	a_{trac}
Wetness	§9.3.7 Surface cond.	Induced wet-surface state	a_{trac}
Precipitation deposits	§9.3.7 Surface cond.	Surface contamination accumulation	a_{trac}
Lateral Stability (a_{stab}) — ISO §10.2.3			
Wind intensity	§10.2.3 Weather	Lateral aerodynamic loading	a_{stab}
Scenario geometry — not ISO-derived			
Ego/adv. start & goal	—	Trajectory intersection topology	Conflict Geometry g
Cut-in direction	—	Agent trajectory (binary)	(retained concrete)

† Sun altitude angle applies to the Vehicle-Cyclist and CutIn scenarios only. Fog density, fog distance, road friction, wetness, and precipitation deposits are CARLA simulator parameters mapped to the ISO §10.3 and §9.3.7 mechanisms, respectively.

III. Evaluation

We evaluate BayScen in CARLA [17] across three functional scenarios from NHTSA's Pre-Crash taxonomy [18]. Each test instance places an ego AV in conflict with an adversarial agent governed by an A* planner and PID controller.

A. Scenarios, Systems Under Test, and Baselines

S1 (Vehicle–Vehicle Junction). Non-signalized junction conflicts between two vehicles (left-turn and right-turn subcases); adversarial vehicle cuts into the ego path from a perpendicular direction.

S2 (Vehicle–Cyclist Junction). Same junction geometry with a pedalcyclist adversary; adds time-of-day as an environmental parameter, capturing lighting sensitivity critical to vulnerable road user perception.

S3 (Vehicle–Vehicle Cut-In). Highway-style lateral encroachment from left or right; exercises a qualitatively different conflict geometry, validating BayScen's ODD abstraction across scenario types.

We assess two contrasting ADS architectures to distinguish system-specific from ODD-level vulnerabilities. **InterFuser** [26]: end-to-end transformer fusing multi-view RGB and LiDAR, top performer on the CARLA 0.9.10 leaderboard. **Modular ADS** [25]: rule-based pipeline (A* planner, PID controller, SSD MobileNet perception), with explicit separation of perception, planning, and control. Each scenario is executed three times per method per SUT.

We compare against six baselines spanning all major paradigms. **Random Sampling**: uniform sampling over all variable combinations; uninformed lower bound. **SitCov** [25]: coverage-weighted sampling prioritizing under-tested values toward near-uniform distribution. **PICT-2w** and **PICT-3w** [6, 27]: pairwise and three-way combinatorial testing ensuring all value pairs or triples appear at least once. **CTBC** [5]: importance-weighted pairwise combinatorial testing; we adapt the original hierarchical weighting scheme by replacing it with data-driven weights derived from collision rates estimated in preliminary simulations, biasing coverage toward high-risk parameter interactions. **AvFuzzer** [12]: genetic adversarial search directly optimizing for collision induction; strongest criticality baseline. Our implementation follows the original paper's GA hyperparameters (population size 12, crossover rate 0.4, mutation rate 0.3) with local fuzzer and random restart as described in [12]. The chromosome encodes *all* environmental parameters alongside conflict geometry, discretized to the same six-level grid as BayScen; AvFuzzer runs for the same budget (N=648 scenarios) as BayScen to ensure a fair comparison. **BayScen-Common**: ablation inverting rarity prioritization to select the most *common* concrete scenarios; BN structure, capability abstraction, and combinatorial coverage are identical to BayScen, directly isolating the contribution of rarity selection. Random, SitCov, and AvFuzzer run a fixed N=648 for both junction and cut-in, while BayScen variants use N=648 for junction and N=216 for cut-in, matching BayScen's full combinatorial suite.

B. Research Questions and Metrics

RQ1 (Criticality). How effectively does BayScen identify safety-critical scenarios? *Metric*: mean collision rate per method, averaged across repetitions.

RQ2 (Physical Plausibility). Do BayScen's scenarios respect real-world physical laws? Collisions under physically impossible conditions are simulation artefacts, not deployment vulnerabilities. *Metric*: physical violation rate — proportion of generated scenarios violating at least one constraint in Table II, derived from the independently developed taxonomy of Hao et al. [28], eliminating post-hoc selection bias. These constraints are never enforced during generation, so the plausibility reported in RQ2 is measured against an external standard, not built into the generator. C1–C2 encode precipitation–surface dependencies; C3a/C3b encode the full friction–wetness relationship across both regimes of [28] Eq. (1); C4 encodes fog density–distance; C5 encodes wind–fog incompatibility; C6 encodes rain–cloud cover. We additionally report the *Clean Critical Rate*: among discovered safety-critical scenarios, the percentage that are also physically plausible (Table V).

RQ3 (Coverage). Does BayScen achieve comprehensive ODD coverage? A method could find realistic failures while missing entire ODD regions. *Metrics*: three TISA metrics [29] — $areaIS$ (test suite diversity), $areaBugs$ (failure-region diversity), $covIS$ (ODD coverage) — purpose-built for black-box AI

TABLE II: Physical plausibility constraints derived from Hao et al. [28]. Parameters normalised to $[0, 100]$ as in CARLA. P: Precipitation; D: PrecipitationDeposits; W: Wetness; F: RoadFriction; G: FogDensity; L: FogDistance; N: WindIntensity; C: Cloudiness.

ID	Physical relationship	Formal rule	Source
C1	Precipitation causes deposits	$P > 20 \Rightarrow D > 0$	[28] §III-B
C2	Precipitation causes wetness	$P > 20 \Rightarrow W > 0$	[28] §III-B
C3a	Wetness reduces friction (low regime)	$W < 40 \Rightarrow F \leq 1 - W/200$	[28] Eq. (1)
C3b	Wetness reduces friction (high regime)	$W \geq 40 \Rightarrow F \leq 0.6$	[28] Eq. (1)
C4	Fog density determines fog distance	$\lvert L - (100 - G)\rvert \leq \epsilon^*$	[28] Eq. (2)
C5	High wind disperses fog	$N \geq 60 \Rightarrow G \leq 40$	[28] §III-B
C6	Rain requires cloud cover	$P > 20 \Rightarrow C > 0$	[28] §III-B

*Tolerance $\epsilon = 10$; from [28] Eq. (2): $L = 100 - G$.

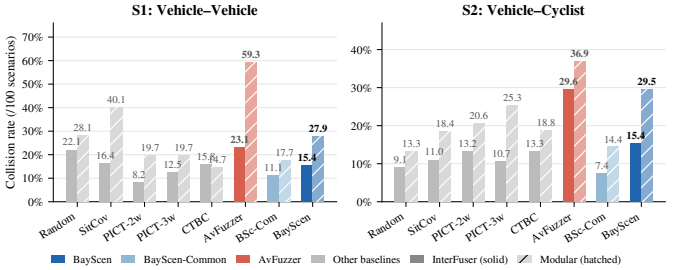


Fig. 4: Collision rate for S1 (Vehicle–Vehicle) and S2 (Vehicle–Cyclist). Solid bars: InterFuser. Hatched bars: Modular. Values annotated on every bar.

system testing and empirically validated to correlate with fault detection.

RQ4 (Failure Characterization). Do capability variables reveal structured failure patterns across degradation conditions? Most methods report only aggregate counts, leaving unanswered *under which conditions* and *why* failures concentrate. *Metric*: collision rate as a function of capability-variable degradation level, separately per SUT and scenario, compared across architectures to surface system-specific versus ODD-level vulnerabilities.

IV. Results

A. RQ1: Criticality

We measure criticality by the *collision rate* (Figure 4, Table III). S3 is omitted: all methods produce near-zero collision rates in the cut-in scenario, consistent with the RQ3 and RQ4 finding that S3 stresses actor-trajectory dynamics rather than environmental degradation.

Three findings structure the RQ1 results.

AvFuzzer leads on raw collision rate — and this is precisely what RQ2 explains. AvFuzzer achieves the highest collision rate in three of four configurations: 23.1%/29.6% for InterFuser and 59.3%/36.9% for Modular (Figure 4, Table III). BayScen is not designed to maximize collision count; it is designed to maximize *realistic* collision count. The gap — 7.7 pp for InterFuser/S1, 14.2 pp for InterFuser/S2 — is the cost BayScen pays for generating collisions under naturalistic environmental conditions. Whether that cost is worth paying is what RQ2 examines: AvFuzzer's additional collisions occur predominantly under physically implausible conditions, which lowers their confidence as evidence of real-world deployment vulnerabilities.

BayScen achieves competitive criticality without adversarial optimization. Random sampling leads BayScen in InterFuser/S1 (22.1% vs. 15.4%), and SitCov leads in Modular/S1 (40.1% vs. 27.9%), both benefiting from large scenario budgets and no realism constraint. BayScen leads or ties the best non-adversarial baseline in InterFuser/S2 (15.4%) and Modular/S2 (29.5%), where the Vehicle–Cyclist geometry concentrates failures in the rare environmental configurations that BayScen systematically targets. The key point is not that BayScen maximizes raw collision count — it does not — but that it achieves competitive criticality while simultaneously providing the coverage and realism guarantees that Random and SitCov cannot (RQ2, RQ3).

TABLE III: Collision rate (%) and mean raw count per run in parentheses (mean over 3 independent runs). N per run: 648 for BayScen, Random, SitCov, AvFuzzer, BayScen-Common; CTBC: 95 (S1), 165 (S2); PICT-2w: 61 (S1), 68 (S2); PICT-3w: 456 (S1), 525 (S2).

	InterFuser		Modular	
Method	S1	S2	S1	S2
Random	22.1 (143)	9.1 (59)	28.1 (182)	13.3 (86)
SitCov	16.4 (106)	11.0 (71)	40.1 (260)	18.4 (119)
PICT-2w	8.2 (5)	13.2 (9)	19.7 (12)	20.6 (14)
PICT-3w	12.5 (57)	10.7 (56)	19.7 (90)	25.3 (133)
CTBC	15.8 (15)	13.3 (22)	14.7 (14)	18.8 (31)
AvFuzzer	**23.1 (150)**	**29.6 (192)**	**59.3 (384)**	**36.9 (239)**
BayScen-Common	11.1 (72)	7.4 (48)	17.7 (115)	14.4 (93)
BayScen	15.4 (100)	15.4 (100)	27.9 (181)	29.5 (191)

Rarity prioritization is a direct contributor to criticality. Comparing BayScen with BayScen-Common isolates the effect of selecting rare versus common concrete scenarios within identical capability configurations. The collision rate increases by **+4.3 pp** (InterFuser/S1), **+8.0 pp** (InterFuser/S2), **+10.2 pp** (Modular/S1), and **+15.1 pp** (Modular/S2) — not a small perturbation: in Modular/S2, rarity prioritization more than *doubles* the collision rate relative to common-scenario selection. Rare driving conditions are disproportionately failure-inducing; BayScen's design exploits this directly, achieving simultaneous improvements in coverage (RQ3) and criticality from the same mechanism.

Statistical significance. Fisher's exact tests on pooled per-run counts (BayScen: $n = 1{,}944$ per group; baselines from $n = 183$ for PICT-2w to $n = 1{,}944$; Benjamini–Hochberg FDR, $\alpha = 0.05$) confirm that BayScen's gains over BayScen-Common are significant in all four S1/S2 conditions ($p < 0.03$ to $p < 10^{-11}$, OR$=$ 1.4–2.8), and over CTBC and PICT-3w in Modular ($p < 0.05$, BH-corrected). In S2, BayScen also significantly outperforms Random and SitCov on both collision and TTC-critical rates ($p < 0.01$, BH). A global Wilcoxon signed-rank test on 18 paired run-level differences (2 SUTs $\times$ 3 scenarios $\times$ 3 runs) confirms consistency for all combinatorial baselines ($W \geq 66$, $p < 0.007$). AvFuzzer leads on raw collision rate by design—a point already established above; RQ2 (Table V) examines the physical plausibility of the scenarios in which those collisions arise.

B. RQ2: Physical Plausibility

Tables IV and V tell a three-part story. Throughout, we treat a physical-constraint violation as a statement about scenario *plausibility*, not collision *causation*: a collision arising in a constraint-violating scenario is not necessarily caused by the violation, but the scenario itself is unlikely to occur in real-world driving, making such findings lower-confidence

TABLE IV: Physical violation rate (%) per constraint and overall (*Any*), averaged across all scenarios and both ADS. *Range*: min–max of *Any* across S1/S2/S3. Mean over 3 independent runs (N per run: 648 for BayScen, Random, SitCov, AvFuzzer; 95/165 for CTBC; 61/68 for PICT-2w; 456/525 for PICT-3w; S1/S2 respectively). Lower is better.

Method	C1	C2	C3a	C3b	C4	C5	C6	*Any*	*Range*
Random	11.0	12.1	2.6	20.5	84.0	25.4	11.0	93.7	92.6–94.8
SitCov	10.5	11.0	2.8	22.4	83.3	25.6	11.5	93.6	93.2–94.0
PICT-2w	10.8	11.9	3.6	21.8	83.5	23.7	11.3	94.3	93.4–95.6
PICT-3w	11.2	10.6	2.4	21.8	82.9	25.3	11.3	93.2	92.5–93.8
CTBC	6.8	7.4	2.4	19.5	79.7	19.4	28.4	93.3	88.4–98.2
AvFuzzer	12.7	10.8	4.0	27.8	82.8	26.8	12.7	94.2	93.0–95.3
BayScen-Common	0.0	0.0	0.0	1.7	0.0	0.8	0.0	2.5	0.5–5.9
BayScen	17.6	10.4	0.0	13.8	5.7	22.6	0.5	52.8	50.9–55.9

TABLE V: Among discovered safety-critical scenarios, the percentage that are also physically plausible (*Clean Critical Rate*, higher is better). Cells: rate % (clean-critical / total-critical), pooled over 3 runs × 2 ADS. S3 excluded: near-zero critical events for all methods.

Method	**S1**	**S2**	**Mean**
Random	8.9 (48/542)	6.2 (11/177)	8.2
SitCov	8.3 (55/662)	9.3 (23/246)	8.6
PICT-2w	18.2 (6/33)	3.4 (1/29)	11.3
PICT-3w	3.8 (8/210)	7.7 (19/246)	5.9
CTBC	2.8 (1/36)	0.0 (0/77)	0.9
AvFuzzer	4.1 (29/709)	7.2 (37/514)	5.4
BayScen-Common	92.3 (275/298)	98.9 (177/179)	94.8
BayScen	**40.5** (176/435)	**39.8** (164/412)	**40.1**

indicators of deployment risk.

The BN eliminates C4 violations entirely, reducing overall violations to 2.5%. BayScen-Common achieves **2.5%** overall violations—a **37× reduction** over all non-BN baselines (93.2–94.3%) and **0%** C4 violations across all scenarios. This is a direct consequence of BN structure learning: the fog density–distance causal relationship is dominant in real meteorological data, and the learned CPDs encode it so strongly that it is preserved in every sampled scenario. Furthermore, **92–99%** of BayScen-Common's critical scenarios are physically plausible (Table V), confirming that the BN architecture alone achieves near-perfect physical plausibility for the configurations it encounters most frequently.

Non-BN methods violate constraints in 93–94% of scenarios; most of their critical findings arise in physically implausible conditions. AvFuzzer—the strongest adversarial baseline—produces physically plausible critical scenarios only **4.1%** (S1) and **7.2%** (S2) of the time. Of 176 critical scenarios found by AvFuzzer in Modular/S1, only 8 are physically plausible under the C1–C6 constraints.

BayScen is 40 pp below the best non-BN baseline and ×7 more likely to surface physically plausible critical scenarios. BayScen reduces violations to 52.8% across all scenarios and both ADS. More importantly, **40.1%** of its critical scenarios are physically plausible, against 5.4% for AvFuzzer and at most 11.3% for all other baselines—the result that matters for AV engineers. The 50 pp gap between BayScen (52.8%) and BayScen-Common (2.5%) quantifies the plausibility cost of rarity prioritization exactly. The per-constraint breakdown explains the mechanism: C4 violations remain near-zero (5.7%) because the fog density–distance dependency is tightly encoded; C3a violations are **0%** because the BN faithfully captures the low-wetness friction regime even under rare sampling; C1, C3b, and C5 rise to 14–23% because rarity sampling stretches softer statistical dependencies toward the tail. This trade-off is structurally inherent to rarity-prioritized sampling—and is precisely what enables BayScen's ODD coverage advantage in RQ3.

To confirm this coverage advantage is not an artifact of the implausible scenarios that rarity sampling admits, we recompute coverage on BayScen's physically plausible subset alone (all of C1–C6 satisfied; only 44–48% of its budget). Even so handicapped to fewer than half its scenarios, BayScen matches or exceeds AvFuzzer's *full*-budget coverage in all four conditions (covIS 50.6–55.1% vs. 47.4–50.5%), so the lead does not depend on physically implausible scenarios. Per-condition before/after coverage is released in the replication package.

Statistical significance. Fisher's exact tests on pooled per-run counts (BayScen: $n = 1{,}944$; baselines from $n = 183$ to $n = 1{,}944$; Benjamini–Hochberg FDR, $\alpha = 0.05$) confirm that BayScen's violation rate is significantly lower than all six non-BN baselines in 36 of 42 comparisons ($p < 10^{-28}$ in every case; global Wilcoxon: $W = 171$, $p < 10^{-4}$, $r = 0.878$). BayScen-Common is the only exception, as expected by design. BayScen's clean critical rate (Table V) is also significantly higher than all non-BN baselines in 22 of 33 per-condition comparisons ($p < 0.05$, BH-corrected; global Wilcoxon: 6 of 7 baselines, $p < 0.012$). The non-significant cases are all in S3, where near-zero critical events leave insufficient power, and in Scenario 1 for PICT-2w, where BayScen-Common's inflated clean rate is the outlier driving the exception.

C. RQ3: Coverage and Diversity

We evaluate test suite quality using the three TISA metrics [29]: $areaIS$ (diversity of the full test suite in ODD feature space), $areaBugs$ (diversity of the failure-inducing region), and $covIS$ (proportion of the feasible ODD explored). Higher values are better on all three. When a method produces *zero failures*, TISA falls back to $areaBugs \equiv areaIS$ [29]; inspection confirms this precisely (difference $< 10^{-3}$) for every zero-failure group. In S3, where most methods find zero or at most one failure, $areaBugs$ is therefore redundant and discussion focuses on $areaIS$ and $covIS$ only. Results are reported in Table VI.

BayScen achieves the highest ODD coverage by a decisive and stable margin. In S1 and S2, BayScen's $covIS$ reaches **62.1–64.3%** across both ADS—a lead of **+12.5 to +16.8 pp** over AvFuzzer, which reaches only 47–51% within its executed budget despite optimising directly for failure discovery; SitCov peaks at 32.0%, Random at 23.9%. Crucially, BayScen achieves this with near-zero variance (std $\leq$ 1.8 pp), whereas AvFuzzer's $covIS$ varies by up to 5.3 pp—a direct consequence of BayScen's structured enumeration covering the same capability configurations every run, regardless of stochastic simulation outcomes.

BayScen surfaces failures across the broadest region of the ODD—not just the densest. In S1, BayScen leads on $areaBugs$ for both ADS: **28.9** vs. 17.1 for InterFuser (×1.69) and **28.5** vs. 21.1 for Modular (×1.35). AvFuzzer's adversarial objective locks it into the highest-return ODD region, producing a failure map that is locally deep but globally narrow. In S2, AvFuzzer leads on $areaBugs$ (35.4 vs.

TABLE VI: TISA coverage metrics (mean over 3 independent repetitions). †In S3 most methods find zero failures, so $aB \equiv aIS$ by TISA's fallback definition and carries no independent information (see text). ⋆S3 suite sizes are design-determined for combinatorial methods: BayScen/BayScen-Common 216, PICT-2w 65, PICT-3w 520, CTBC 739; budget-parameterised methods (Random, SitCov, AvFuzzer) ran a fixed 648 each. AvFuzzer's S3 $covIS$ (48.4 / 51.8%) is therefore at 3× BayScen's executions; CTBC's (64.2 / 61.6%) at 3.4×. Statistical significance of BayScen's advantage is reported in the text (global Wilcoxon signed-rank, 18 paired differences).

	InterFuser									**Modular**								
	S1			S2			S3†⋆			S1			S2			S3†⋆		
Method	aIS	aB	cov	aIS	aB	cov	aIS	aB	cov	aIS	aB	cov	aIS	aB	cov	aIS	aB	cov
Random	6.8	6.7	23.9	12.6	12.5	21.8	15.3	15.3	43.8	2.7	2.7	9.9	12.7	12.4	22.7	13.1	13.1	46.7
SitCov	8.8	8.6	32.0	10.3	10.2	23.7	14.8	14.7	46.9	8.7	7.3	31.8	9.7	9.6	22.8	15.4	15.4	46.4
PICT-2w	2.5	2.0	9.9	6.9	6.9	13.5	7.7	7.7	23.8	3.3	1.8	14.3	5.3	5.2	10.4	7.8	7.8	23.8
PICT-3w	1.8	1.6	10.5	13.1	13.0	23.5	13.2	13.2	41.0	1.9	1.6	10.5	10.3	9.5	18.6	13.2	13.2	41.0
CTBC	7.9	8.2	20.5	26.9	26.0	40.9	0.4	0.4	**64.2**	8.3	7.3	21.3	24.8	23.9	37.9	0.4	0.4	**61.6**
AvFuzzer	17.7	17.1	48.6	33.1	**35.4**	47.4	23.2	23.2	48.4	23.3	21.1	50.5	**46.9**	**44.7**	47.5	**23.6**	**23.6**	51.8
BayScen-Common	0.0	0.0	0.0	0.0	0.0	0.0	1.1	1.1	47.9	0.0	0.0	0.0	0.0	0.0	0.0	1.1	1.1	47.9
BayScen	**29.4**	**28.9**	**62.1**	**34.5**	33.8	**64.1**	**23.2**	**23.7**	49.1	**28.7**	**28.5**	**63.0**	37.8	37.1	**64.3**	23.3	23.3	48.6

33.8 for InterFuser; 44.7 vs. 37.1 for Modular) and on $areaIS$ for Modular (46.9 vs. 37.8): the Vehicle–Cyclist geometry contains a high-failure-density region that adversarial search exploits efficiently. Yet even in S2, BayScen achieves the highest $covIS$ (64.1%/64.3%) and the highest $areaIS$ for InterFuser (34.5), confirming that broader ODD coverage is maintained even when local failure density is not maximized.

The BayScen-Common ablation: a definitive isolation

BayScen-Common—identical to BayScen except that it selects the *most common* concrete scenarios—achieves $areaIS = areaBugs = covIS = 0$ in both S1 and S2 for both ADS. This is not a computation artefact: **98.9%** of BayScen-Common scenarios have zero fog, zero precipitation, and zero road wetness; **83.3%** fall into just three concrete parameter combinations, with the single most frequent accounting for **49.4%** of the entire suite. Despite enumerating all 648 capability configurations, BayScen-Common places all of them at nearly the same point in ODD feature space — TISA measures convex-hull area, which collapses to zero when 83% of points are geometrically coincident.

This exposes the *coverage paradox* of naturalistic data: common driving conditions are not diverse ones. Every advantage BayScen holds over every other method on every TISA metric in S1 and S2 is driven primarily by the rarity prioritization step. In S3, BayScen-Common partially recovers ($areaIS$ = 1.1, $covIS$ = 47.9) because cut-in configurations are more uniformly distributed in naturalistic data, confirming the mechanism: the S1/S2 collapse is the direct consequence of natural data's concentration in benign junction conditions.

S3 defines the scope of applicability

In S3, suite sizes are design-determined and heterogeneous (see Table VI footnote), so raw $covIS$ values are not directly comparable at face value. *Against AvFuzzer:* at equal executed-scenario budgets (both at 216 scenarios) BayScen leads on $covIS$ by **+13.9 pp** (InterFuser) and **+10.1 pp** (Modular); the wall-clock-matched comparison (Table VII) further confirms a **+12.5 pp** and **+6.4 pp** advantage after crediting AvFuzzer the full generation overhead. BayScen thus achieves comparable or superior coverage with one-third the executions of budget-parameterised baselines. *Against CTBC:* its raw $covIS$ advantage (64.2%/61.6% vs. BayScen's 49.1%/48.6%) accrues at 739 executions—3.4× BayScen's 216—and a like-for-like budget comparison is unavailable because CTBC's covering-array suite size is fixed by design and cannot be truncated. AvFuzzer leads on $areaIS$ (23.2/23.6 vs. BayScen's 23.2/23.3—margins within 1.5%); BayScen is competitive on $areaIS$ but does not dominate. Consistent with the RQ4 finding that S3 stresses actor-trajectory dynamics rather than environmental capability degradation, the capability abstraction provides less differentiation when the failure-driving variables fall outside its scope—a boundary that precisely characterises BayScen's domain of applicability.

Equal-budget comparison. BayScen and BayScen-Common require an offline scenario-generation phase that AvFuzzer—which generates scenarios online during execution—does not. To ensure fairness we evaluate under (i) *equal executed-scenario budgets* and (ii) *equal wall-clock budgets*. For S1 and S2 all methods share $N = 648$, so regime (i) is unambiguous and results are in Table VI. For S3, suite sizes are determined by combinatorial design and are inherently unequal: BayScen exhausts its capability space at 216 scenarios while CTBC ran 739 and PICT-3w ran 520 (both more than BayScen by design), and budget-parameterised methods (Random, SitCov, AvFuzzer) ran a fixed 648. Because these suite sizes cannot be equated, equal-executed and wall-clock-matched comparisons for S3 are reported jointly in Table VII. For the wall-clock criterion we convert BayScen's offline generation time into its AvFuzzer equivalent: the number of additional scenarios AvFuzzer could execute in the same wall-clock time. Using per-scenario AvFuzzer execution times measured from the raw simulation logs (107–201 s/scenario for InterFuser; 27–48 s/scenario for Modular, varying by scenario type), BayScen's generation overhead ($\leq$1h52m) translates to 16–38 extra AvFuzzer scenarios on InterFuser and 80–139 on Modular, with matched counts of 241 (S3, InterFuser) to 787 (S1, Modular). $covIS$ at matched counts is read from cumulative TISA learning curves (prefix sizes $k \in \{50, 100, 200, 400, N_{\text{base}}\}$); values above $k = 648$ are linearly extrapolated from the 400→648 slope, and AvFuzzer's equal-executed $covIS$ for S3 is interpolated at $k = 216$.

Under both budget regimes BayScen leads AvFuzzer in every configuration (Table VII): **+10.1 to +16.8 pp** equal-executed and **+6.4 to +16.6 pp** equal-wall-clock. The worst case is Modular/S3, where the short generation time (≈45 min) and fast per-scenario execution (27 s) let AvFuzzer run 100 extra scenarios, narrowing the gap to 6.4 pp—still decisive. BayScen's coverage advantage is thus robust to both budget accounting and generation overhead.

TABLE VII: Budget comparison under equal executed-scenario and equal wall-clock budgets (BayScen only; see text for BayScen-Common). *Gen*: offline generation time. *Extra*: additional AvFuzzer scenarios executable in BayScen's generation time. *Match.*: N_{base} + Extra. *covIS* are means over 3 runs. [a]Equal-executed AvFuzzer *covIS* equals Table VI for S1/S2 ($N = 648$); for S3, interpolated at $k = 216$. †interpolated; ‡linearly extrapolated beyond $k = 648$.

			BayScen	Equal executed (N_{base})		Equal wall-clock (matched)	
Config	**Gen**	**Extra**	**Match. cov @ N_b**	**AF cov @ N_b**	**Gap**	**AF cov @ match.**	**Gap**
InterFuser/S1	1h52m	38	686	62.1%	48.6%[a]	+13.5 pp	49.3%‡	**+12.8 pp**
InterFuser/S2	0h55m	16	664	64.1%	47.4%[a]	+16.7 pp	47.5%‡	**+16.6 pp**
InterFuser/S3	0h45m	25	241	49.1%	35.3%†	+13.9 pp	36.6%†	**+12.5 pp**
Modular/S1	1h52m	139	787	63.0%	50.5%[a]	+12.5 pp	54.3%‡	**+8.8 pp**
Modular/S2	0h55m	80	728	64.3%	47.5%[a]	+16.8 pp	49.9%‡	**+14.5 pp**
Modular/S3	0h45m	100	316	48.6%	38.4%†	+10.1 pp	42.1%†	**+6.4 pp**

TABLE VIII: Collision rate (%) at minimum (L0) and maximum (L4) degradation per capability variable. BayScen only; S1 values; each cell is the mean over 3 runs, and Δ is the L4−L0 change (positive Δ indicates degradation increases collisions).

Capability	**ADS**	**L0**	**L4**	**Δ**
Sensor Perception (a_{perc})	InterFuser	2.4	21.1	+18.7
	Modular	13.8	25.6	+11.8
Surface Traction (a_{trac})	InterFuser	6.5	9.8	+3.3
	Modular	12.7	25.5	+12.8
Lateral Stability (a_{stab})	InterFuser	8.6	7.6	−1.0
	Modular	17.2	14.4	−2.8

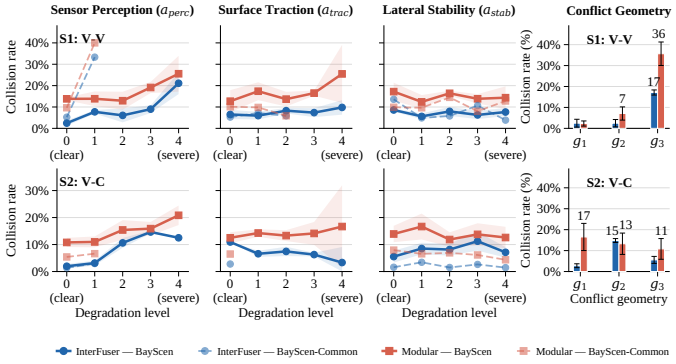


Fig. 5: Collision rate as a function of capability-variable degradation level (0 = clear; 4 = severe) for S1 (top row) and S2 (bottom row). Solid lines with ±1 std bands: BayScen. Dashed lines: BayScen-Common (plotted only at levels reachable by common-scenario selection). Right column: collision rate per conflict-geometry state (g_1, g_2, g_3), S1 and S2 on a shared y-scale.

Statistical significance. Global Wilcoxon signed-rank tests on 18 paired run-level differences (2 SUTs × 3 scenarios × 3 runs per baseline) confirm BayScen's dominance: 19 of 21 (baseline × metric) comparisons are significant ($p < 0.001$, effect size $r = 0.878$ in all but one case). The two non-significant cases are BayScen vs. AvFuzzer on *areaIS* and *areaBugs* ($p \approx 0.28$ and 0.23), consistent with AvFuzzer's competitive S2 performance noted above. BayScen significantly outperforms AvFuzzer on *covIS* ($p = 0.001$, two-sided; $r = 0.72$), confirming that broader ODD coverage—BayScen's primary claim—holds even against the strongest adversarial competitor.

D. RQ4: Failure Characterization

For each capability variable and degradation level (0 = undegraded; 4 = maximally degraded), we compute the mean collision rate over three repetitions for BayScen and for the BayScen-Common ablation. BayScen-Common keeps capability coverage identical but selects the *most common* concrete scenarios; its dashed lines in Figure 5 therefore stop wherever common driving data runs out—those absent levels are structurally unreachable by common-scenario selection, not missing by accident. S3 is excluded: both methods produce near-zero rates at all levels, confirming that cut-in conflicts stress actor-trajectory dynamics rather than environmental capability degradation.

The profiles in Figure 5 and Table VIII tell a coherent, three-part story about where, why, and for whom SUT failures concentrate—none of which aggregate collision counts reveal.

Perception degrades both architectures, but at radically different rates. At clear conditions (level 0), InterFuser's multi-modal fusion gives it a ×5.8 safety advantage over the Modular pipeline (2.4% vs. 13.8%). That advantage erodes monotonically as optical degradation intensifies, collapsing to a mere 4.5 pp gap at level 4 (21.1% vs. 25.6%)—a deployment ceiling beyond which ADS-level engineering cannot compensate for physical sensor occlusion. Traction behaves oppositely: InterFuser is essentially immune across all surface conditions (6.5%–9.8%), while Modular nearly doubles its failure rate from 12.7% to 25.5% as traction degrades. These two variables partition the failure landscape cleanly: perception is a shared vulnerability gated by severity, traction is an architecture-specific weakness exclusive to Modular. Neither pattern is accessible to BayScen-Common—it covers only levels 0–1 for a_{perc} and misses levels 3–4 for a_{trac} in S1 and levels 1–4 in S2, structurally concealing both the severity-gated convergence and the full extent of Modular's traction escalation.

Lateral stability is an ODD non-issue, and proving it matters. Both architectures show flat, level-invariant rates under a_{stab} (InterFuser: 5.6%–8.6%; Modular: 12.4%–17.2%). This is not a null result: it is actionable negative evidence. No existing method can distinguish "a_{stab} does not matter" from "a_{stab} was never tested under sufficient severity"—because no existing method reaches all degradation levels by construction. BayScen does, and the flat profile across all five levels is precisely what allows this conclusion to be stated with confidence: AV validation engineers can deprioritize wind-load test cases without sacrificing safety coverage.

Conflict geometry undergoes a complete inversion between actor types, and BayScen-Common is blind to it. In S1 (Vehicle–Vehicle), both architectures concentrate failures at g_3: 17.1% for InterFuser, 35.6% for Modular, while g_1 and g_2 remain below 7%. Switching to S2 (Vehicle–Cyclist), this distribution inverts entirely: g_3 collapses to 5.6% for InterFuser as g_2 surges from 2.5% to 14.8%—a ×6 rise from a configuration that was near-zero in S1. The inversion is geometric, not stochastic: the cyclist's narrower lateral profile eliminates the perpendicular-approach conflict that makes g_3 dangerous in vehicle–vehicle interactions and instead exposes a crossing-path vulnerability unique to g_2. BayScen-Common assigns exactly 0% of its S2 budget to g_3—the geometry responsible for 35.6% of S1 Modular failures—not by design, but because rare conflict configurations are absent from common driving data by definition. The shared y-scale in Figure 5 (right column) makes this reversal immediately

visible.

Abstraction Validation

The three capability variables were defined once from ISO 34503 and applied unchanged to S1, S2, and S3. If the abstraction were uninformative, rates would be flat and undifferentiated across levels and identical across scenarios. Instead, each variable reveals a distinct, reproducible signal—severity-gated convergence, architecture-discriminating escalation, scenario-specific geometric inversion—that holds consistently across both ADS and both scenarios. The abstraction is not a modelling convenience; it is a diagnostic instrument that transforms a test suite into a structured map of ADS vulnerability, exposing patterns that are invisible to any method reporting only aggregate collision counts. The abstraction's primary contribution is this diagnostic structuring (RQ4); tractable coverage (RQ3) is the mechanism that enables it—ensuring every capability level is exercised so that the failure profile is complete, not sparse.

V. Related Work

A. Scenario-Based Testing

Scenario-based testing decomposes the ODD into structured, reproducible test cases [2, 30]. The standard taxonomy distinguishes functional, logical, and concrete scenarios [2]; the transition from logical to concrete creates a combinatorial explosion that grows exponentially with parameter count [31, 32], motivating the coverage–realism–criticality trilemma BayScen navigates.

B. Scenario Generation Methods

Search-Based Methods treat generation as an optimization problem. Boundary-focused approaches use Gaussian Processes [9, 33, 34, 35]; evolutionary algorithms [12, 36, 37] and RL [7, 8, 38, 39] directly maximize failure induction. While effective at finding critical scenarios, these methods provide no coverage guarantees and frequently violate naturalistic driving dependencies [10, 11]. Recent realism-aware adversarial frameworks—KING [40], AdvDiffuser [41], and SaFeR [42]—constrain optimization to behaviorally plausible domains, but operate at the level of *dynamic element interactions* (actor trajectories). BayScen operates at the orthogonal level of *environmental condition degradation* (weather, lighting, surface friction) as defined by ISO 34503; direct comparison is therefore infeasible by design.

Combinatorial Testing covers parameter interactions systematically [43], including CTBC [5], PICT-based 3-way testing [6], and feature model sampling [44]. These methods guarantee structural coverage but treat all parameter combinations as equally probable, wasting budget on scenarios that could never occur in real-world driving.

Data-Driven Methods extract scenarios from naturalistic data [15, 45, 46], accident databases [47, 48], and sensor logs [49] via clustering, pattern mining, and kernel density estimation [13, 50, 51]. Importance sampling [52, 53, 54] and risk-constrained GMMs [14] accelerate rare-event discovery, but these methods are fundamentally trapped by a coverage paradox: they can only reproduce scenarios resembling their training distribution, systematically underrepresenting the rare tail that matters most for safety [14].

Generative AI Methods have recently emerged as a powerful paradigm. Diffusion models generate controllable, realistic traffic scenarios [41, 55]; LLM-based frameworks such as ChatScene [56] translate natural language into executable scenarios; and multi-modal approaches integrate textual and visual inputs [57, 58]. While these methods produce diverse and behaviorally realistic outputs, they inherit their training data's limitations: they struggle to systematically explore the extreme ODD tail without hallucinating physically implausible conditions, and provide no coverage guarantees.

Knowledge-Based Methods use expert rules, ontologies, and formal specifications [59, 60, 61]. They ensure broad ODD coverage but rely on exhaustive manual specification and may miss failure modes unanticipated at design time.

C. Positioning BayScen

Each paradigm optimizes one dimension at the cost of the others: search-based methods maximize criticality but sacrifice realism and coverage; data-driven methods preserve realism but cannot reach the dangerous tail; combinatorial methods cover the parameter space but treat realistic and implausible scenarios alike; generative AI methods produce diverse outputs but provide no structural coverage guarantee. BayScen navigates this trilemma by integrating a Bayesian Network (realism), ISO 34503-grounded capability abstraction (tractable coverage), and rarity-prioritized sampling (criticality)—a combination not achieved by any prior work.

Prior pairings of Bayesian Networks with combinatorial testing for AV scenario generation (e.g., [62]) operate over raw scenario parameters: the BN models traffic-participant relationships, while hand-built constraint sets (forbidden tuples) prune invalid combinations before t-wise coverage is applied. Such constraints *filter* the parameter space without *redefining* it—coverage remains over raw combinations and the suite carries no capability-level diagnostic structure. BayScen's capability-abstraction layer instead redefines "coverage" from raw combinations to functional degradation regimes, so that BN sampling and combinatorial coverage operate at a resolution that is simultaneously tractable and diagnostically informative.

VI. Threats to Validity

Construct validity. Physical plausibility uses the independently developed constraint taxonomy of Hao et al. [28], which has no connection to our work and so eliminates post-hoc selection bias; no constraint set is exhaustive, but an external framework is the strongest available mitigation. ODD coverage uses the TISA metrics [29], purpose-built for black-box AI testing and validated to correlate with fault detection.

Abstraction bias. If the three capability variables collapsed behaviorally distinct combinations, the abstraction would produce coverage gaps. We address this at three levels: (i) the groupings follow the ISO 34503:2023 attribute taxonomy (§9.3.7, §10.2.3, §10.3, §10.4) and were fixed before any experimental results—a design choice, not a post-hoc rationalization; (ii) RQ4 empirically validates it—were the variables uninformative, collision rates would be flat; instead each reveals a distinct, reproducible signal (severity-gated

convergence, architecture-discriminating escalation, geometric inversion) across both ADS and all scenarios; (iii) the conflict geometry variable reduces 81 trajectory combinations to 3 topologically distinct states, confirmed by the RQ4 inversion (g_3 dominant in S1, g_2 surging $\times 6$ in S2).

BN mis-specification. Structural errors are mitigated by the error-tolerant knowledge-guided learning of Ban et al. [20], which uses domain knowledge as soft constraints overridable by data; parametric errors by BDeu smoothing (ESS $= 5$). RQ2 provides direct empirical validation: BayScen-Common achieves near-zero violation rates, confirming that real-world physical dependencies are faithfully encoded in the learned CPDs. We further assess structural reliability by data-only bootstrap resampling ($B = 100$, soft priors removed to isolate empirical support): 9 of the 11 structural edges recur in $\geq$90% of resamples. Of the two edges below this threshold, one is recovered only inconsistently by the data—a weaker data-driven dependency—while the other is supplied by the soft priors, encoding a physical relationship the data alone does not exhibit; the priors are thus decisive for a single edge, leaving the structural backbone data-driven. As a generalization check, 5-fold cross-validation over all 43,207 observations shows the learned BN improves held-out log-likelihood by $+1.92$ nats/sample over an independence (product-of-marginals) baseline (from -7.39 to -5.47 for S1 and from -9.29 to -7.36 for S2/S3), a gain stable across all folds (std < 0.02); the conditional dependencies therefore capture genuine co-occurrence rather than overfitting.

Simulation, stochasticity, and geographic bias. CARLA may not fully reproduce real-world sensor and physics complexity, but this affects all methods equally, preserving relative comparisons. Each scenario is executed three times with mean rates and std bands reported; three repetitions is a known limitation for low collision rates, partially offset by margins consistent across both ADS. The BN is learned from Norwegian meteorological data, but encodes geographically invariant causal relationships, limiting the impact on generated-scenario realism.

Scenario and ADS scope. Evaluation covers three NHTSA scenarios, two conflict geometries, two actor types, and two contrasting ADS architectures. Generalisation to pedestrian interactions, highway merging, or unstructured environments requires further validation; the ODD-level capability variables are environmental and transfer unchanged, while the intersection-specific conflict-geometry component must be re-derived per conflict type via the same topological-reduction principle [25]. The consistency of BayScen's advantages across both ADS architectures supports generalizability beyond any single implementation.

VII. Conclusion

BayScen navigates the coverage–realism–criticality trilemma in scenario-based AV testing by abstracting environmental parameters at the level of the safety-critical capabilities they degrade—grounded in ISO 34503:2023—and coupling this abstraction with a Bayesian Network learned from real-world meteorological data and rarity-prioritized combinatorial enumeration. Across three NHTSA scenarios, two ADS architectures, and six baselines, BayScen's critical scenarios are $\times 7$ more likely to be physically plausible than those of the strongest adversarial baseline (40% vs. 4–7%), and it leads all methods on ODD coverage by 12–17 percentage points; capability-organized failure profiling further reveals architecture-specific vulnerability patterns — perception convergence under extreme degradation, traction escalation exclusive to the modular pipeline, geometric inversion between actor types — invisible to methods reporting only collision counts. The practical implication is direct: the question is not how many collisions a suite finds, but how many represent vulnerabilities likely to manifest under realistic deployment conditions—which BayScen is designed to maximize.

Future Work

The primary limitation is the 52.8% physical-violation rate under rarity-prioritized sampling—rarity and physical implausibility are partially co-selected by design, with BDeu smoothing (ESS = 5) possibly inflating mass on implausible combinations—so reducing violations while preserving tail-focused selection is the main avenue for future work, alongside extending the ISO 34503-grounded capability derivation to dynamic and scenery elements and coupling BayScen with realism-aware adversarial trajectory frameworks (KING [40], AdvDiffuser [41], SaFeR [42]) for joint worst-case environmental and actor-behavior discovery.

Data Availability

The BayScen implementation, scenario sets, and evaluation scripts are available at https://doi.org/10.5281/zenodo.19748457.

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